

Fuzzy Regression Analysis and Application to a Crisp Model

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1. Introduction

When modelling real processes, one is often faced with vague non-stochastic data due to *incomplete information* (e.g., interval values, $x > a$, $x < b$, $a < x < b$), *inaccuracy* (e.g. non-measurable variables), *linguistic vagueness* (evaluation of expressions like "is present", "barely present", "unimportant", "important"; score scales). Here, modelling in a fuzzy environment will be considered in a traditional way, i.e. from observations (data) we build up a model being optimal in a certain sense and describing the relationship between independent and dependent variables. Fuzzy Sets Theory turns out to be the appropriate apparatus for processing vague and fuzzy information [1-3]. In the remaining part of this section we give some elementary fuzzy sets fact.

Definition 1. Let U be a crisp set. Moreover, let A be any fuzzy set over U with membership function μ .

Definition 2. If $U = \mathbb{R}^1$ (real axis), then A is a *fuzzy number* iff

1. $\exists$ at least one x^* with $\mu_A(x^*) = 1$.
2. μ_A is upper semicontinuous and quasi-concave.
3. All level sets A_α are bounded for $\alpha \in (0,1]$.

Definition 3. Fuzzy number A is of *symmetric L-type* if there are real numbers b , $s(>0)$ and a function L with

1. $L : [0,\infty) \rightarrow [0,1]$; $L(0) = 1$, $L(\infty) = 0$.
2. L is continuous, strict monotonous decreasing and

$$\mu_A(t) = L\left(\frac{|t-b|}{s}\right). \quad (1)$$

Here, b is the *mean value* and s is the *spread*. (For $s = 0$ we get the formal representation of

crisp numbers).

In the following we will assume that the input-output model is parametric and crisp whereas the observations are fuzzy numbers (resp. vectors of the latter). Thus we are led to a modification of fuzzy regression analysis [2].

2. Technical background

In the Moscow Pipe Company "FILIT" in welding aggregates, cooling lubricants (*CL*) are used in an amount of 40-50 m^3 . These *CL* are mixtures of water and mineral technical oil based emulsion. The higher the emulsion's concentration in water the better its corrosion and biological properties and the longer the pipes are protected against corrosion. However, unreasonable increase of concentration will lead to unnecessary consumption of emulsion and hence, to economic loss.

In the plant, quality control is carried out by measuring the emulsion's concentration (%) in *CL*, corrosivity, and biological impairments. Measurements turn out to be fuzzy.

Concentration is measured by an *indirect weighting method* (which is adulterated by destructive products and leaks). Corrosivity and biological impairments are evaluated by *score scales*.

The evaluation of corrosivity is performed by the degree of corrosion of cast iron chips, i.e.

no corosion	— 0 points
traces of c.	— 1 p.
weak c.	— 2 p.
moderate c.	— 3 p.
strong c.	— 4 p.

Biological impairments are evaluated through the amount of microorganisms per unit volume (characterized by smell and intensity of colouring when bringing in a special substance). This leads to the following table.

colour	smell	evaluation (points)
white	none	0
light pink	none	1
pink	none	2
loud pink	weak	3
red	weak	4
raspberry	pungent	5

CL is usable for biological impairment less than 5, otherwise it will be removed from the welding aggregate. If corrosivity of *CL* exceeds 1, then it is unusable for further welding.

Now, the technical problem consists in finding the minimum concentration of emulsion ensuring corrosivity ≤ 1 and biological impairment ≤ 4 .

Inputs (independent) variables are given by concentration (x_1) and impairment (x_2). The *output* variable y is the corrosivity. Deeper analysis shows a linear connection between inputs and output, i.e.

$$y(\mathbf{x}, \mathbf{a}) = a_0 + a_1 x_1 + a_2 x_2 \quad (2)$$

with $\mathbf{x} = (x_1, x_2)$ and suited parameter vector $\mathbf{a} = (a_0, a_1, a_2)$.

Now we have to solve the following tasks:

1. Determine $\mathbf{a}$ from given fuzzy information
2. Using the model obtained, solve (for positive x_1)

$$x_1 \rightarrow \min \quad (3)$$

$$0 \leq x_2 \leq 4$$

$$y \leq 1$$

3. Fuzzy regression - the problem

We will perform modelling under the following assumptions.

1. The relationship between inputs and output is described by a crisp linear parametric model with parameter vector $\mathbf{a} \in \mathbb{R}^k$.
2. Experimental inputs $\mathbf{X}_i = (X_{i1}, \dots, X_{in_i})$ and outputs Y_i are given as fuzzy numbers with memberships $\{\mu_{X_{ij}}\}$ and μ_{Y_i} . Here, $i = 1, \dots, N$ (number of experiments), $j = 1, \dots, n$ (dimension of input).

The Problem.

By the experimental fuzzy data the optimal (in a certain sense) parameter vector $\mathbf{a}^*$ is to be determined.

Definition 4. Let $\mathbf{x}_i \in \mathbb{R}^n$ and y_i real; $i = 1, \dots, N$. Then fuzzy set $\mathbf{Q}_i = (\mathbf{X}_i, Y_i)$ is called *fuzzy point* if

$$\mu_{\mathbf{Q}_i}(\mathbf{x}_i, y) = \mu_{X_{i1}}(x_1) \cdot \dots \cdot \mu_{X_{in_i}}(x_n) \cdot \mu_{Y_i}(y) \quad (4)$$

Definition 5 [2]. Let $\mathbf{x}_i \in \mathbb{R}^n$, $\mathbf{a} \in \mathbb{R}^k$, and be $y = f(\mathbf{x}, \mathbf{a})$ a crisp function. For given fuzzy point $\mathbf{Q}_i$ we introduce the *similarity measure* of $\mathbf{Q}_i$ with function f (for vector $\mathbf{a}$) by

$$M_i(\mathbf{a}) = \sup_{\mathbf{x} \in \mathbb{R}^n} \mu_{\mathbf{Q}_i}(\mathbf{x}, f(\mathbf{x}, \mathbf{a})). \quad (5)$$

We can interpret M_i as measure of compatibility of point $\mathbf{Q}_i$ with the graph of f .

Now suppose to be given N fuzzy points $\{\mathbf{Q}_i\}$. We introduce the *aggregation operators* $MA(\mathbf{a})$ and $MG(\mathbf{a})$ as

$$MA(\mathbf{a}) = \frac{1}{N} \cdot \sum_i M_i(\mathbf{a}) \text{ (arithmetic mean),} \quad (6)$$

$$MG(\mathbf{a}) = \left(\prod_i M_i(\mathbf{a}) \right)^{1/N} \text{ (geometric mean).} \quad (7)$$

Recall the following properties of each. Whereas MA evaluates the "mean crossing" of f through all Q_i (compensatory), MG is an evaluation for the "simultaneous crossing" of f through all Q_i (little compensatory effect leading to weakest link policy).

The *optimal parameter vector* $\mathbf{a}^*$ is determined from

$$M\#(\mathbf{a}^*) = \max_{\mathbf{a} \in \mathbb{R}^k} M\#(\mathbf{a}) \quad (8)$$

with $\# = A, G$.

Correspondingly, the *optimal model* sounds as

$$y = f(\mathbf{x}, \mathbf{a}^*) . \quad (9)$$

However, one should be cautious when choosing the aggregation operator. Non-compensatory evaluation is problematic, since outliers can lead to false models !

4. Fuzzy regression - the Solution

Suppose inputs and outputs to be modelled by L -type fuzzy numbers, i.e.

$$\mu(t) = \exp \left\{ - \left(\frac{|t-b|}{s} \right)^m \right\}; \quad s \geq 0, m > 1. \quad (10)$$

In Fig. 1 we depicted the case $b = 0, s = 1$ for different m .

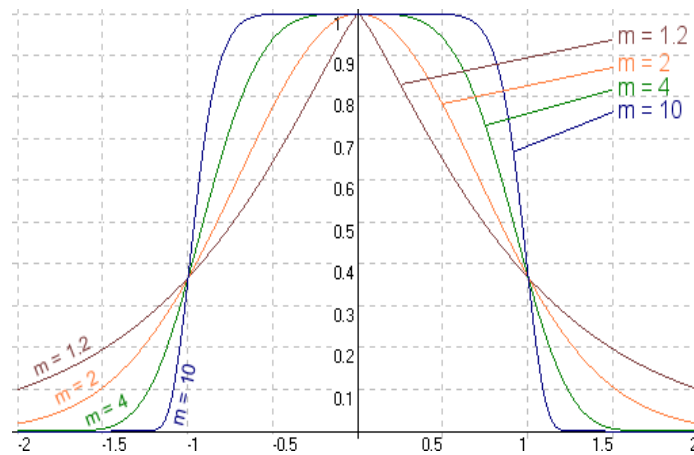


Fig. 1. Family of membership functions for fuzzy (L,L) -numbers in the sense of Dubois and Prade

The result of measurement is given by b , the spread (fuzziness) can be modelled by s . By va-

rying m one can reach further adaption.

$m \approx 1$: Mean value b is more or less known, but there is little fuzziness or it cannot be evaluated.

$m \approx 2$: Mean value and fuzziness can be evaluated.

$m \approx 10$: Fuzzy number of interval type.

Thus, we have

$$\mu_{x_{ji}}(t) = \exp \left\{ - \left(\frac{t - x_{ji}}{\beta_{ji}} \right)^m \right\}; \quad \mu_{y_i}(t) = \exp \left\{ - \left(\frac{t - y_i}{\alpha_i} \right)^m \right\}. \quad (11)$$

For the linear model we obtain

$$M_i(\mathbf{a}) = \exp(-d_i(\mathbf{a})) \quad (12)$$

with

$$d_i(\mathbf{a}) = \frac{\left| \left(a_0 + \sum_j^n a_j \cdot x_{ji} \right) - y_i \right|^m}{\left(\alpha_i^{m \cdot m-1} + \sum_j^n |a_j \cdot \beta_{ji}|^{m \cdot m-1} \right)^{m-1}}. \quad (13)$$

Hence

$$\mathbf{a}_A^* = \arg \max_{\mathbf{a} \in \mathbb{R}^{n+1}} \sum_i e^{-d_i(\mathbf{a})}, \quad \mathbf{a}_G^* = \arg \min_{\mathbf{a} \in \mathbb{R}^{n+1}} \sum_i d_i(\mathbf{a}). \quad (14)$$

When solving (14) we are faced with numerical difficulties which are due to the high nonlinearity implying poly-extremality. Here, a combination of Random Search Methods and Davidson-Fletcher-Powell Procedure (DFP) turned out to be best [4,5].

The algorithm of linear regression analysis has been realized in C++ in a WINDOWS 95 environment. Besides the optimal solutions (14) and corresponding $MA(\mathbf{a}^*)$ и $MG(\mathbf{a}^*)$ we can obtain the least squares solution $\mathbf{a}^{LS}$, and for each experimental point Q_i we determine its individual evaluation $M_i(\mathbf{a}^*)$ thus detecting outliers.

The program also makes possible the visualization for one-dimensional regression. In Fig. 2 shows the situation for the model $y = 2 + 2x$ and $m = 1.2$. The behavior of the aggregation operator is depicted in Fig. 3, whereas in Fig. 4 the confidence set

$$D(\mathbf{a}) = \{\mathbf{a}: \mathbf{a} = \arg\{MA(\mathbf{a}) \geq MA(\mathbf{a}^*) - \delta\}$$

for fixed positive δ is given.

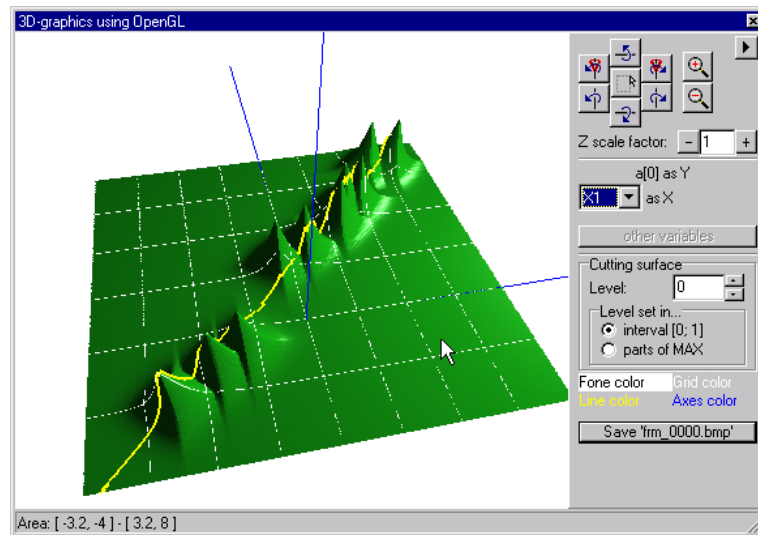


Fig.2. Linear model fitting fuzzy points.

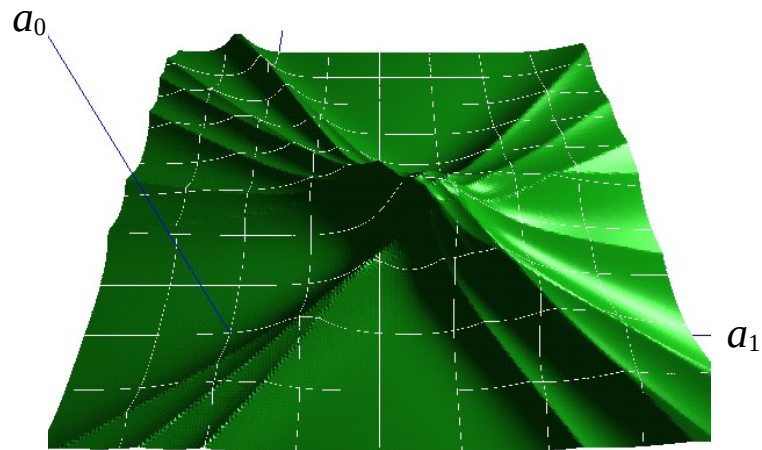


Fig. 3. Aggregation operator "arithmetic mean" in the parameter space.

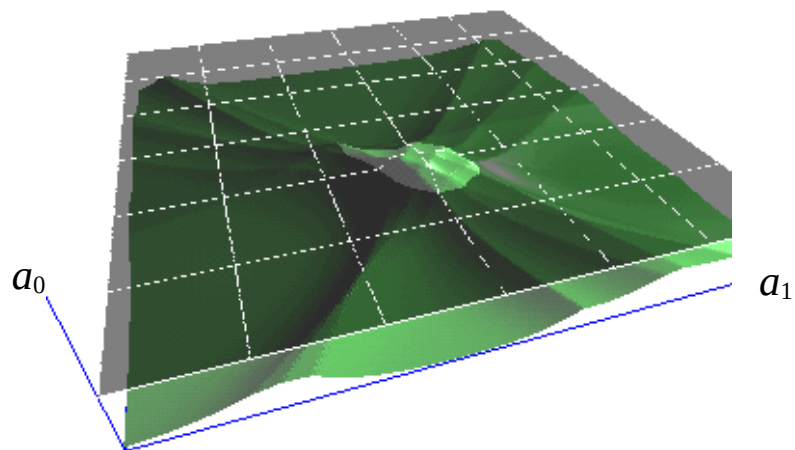


Fig. 4. Area with given value of aggregation operator in the parameter space.

5. Solution analysis

The vagueness of input information leads to vagueness when determining m , α_i , β_{ji} . Among others we are faced with the problem of stability of the optimal solution $\mathbf{a}^*$ with respect to variations of m , α_i , β_{ji} and to outliers, as well. Moreover, it is salutary to compare $\mathbf{a}^*$ with the least squares solution $\mathbf{a}^{LS}$.

We analyzed different models and cite one of them for illustration.

Example. Consider $y = 5 + 6x_1 - 3x_2 - x_3 + \zeta$ where ζ is a random noise with density

$$p(u) = \begin{cases} (2.5 / \varepsilon^5) \cdot u^4 & \text{for } -\varepsilon \leq x \leq \varepsilon, \\ 0 & \text{otherwise} \end{cases}$$

for given positive ε ($= 1$). We used $N = 16$ data points; $m = 1.2, 2, 4, 10$; $\alpha_i = \alpha \in \{0.1, 1, 5, 10\}$; $\beta_{ji} = \beta \in \{0.025, 0.25, 1\}$. In Table 1 we gathered the corresponding results.

Table 1. Optimal parameters

Operator	MA				MG			
m	1.2	2	4	10	1.2	2	4	10
$\mathbf{a}^*$ (true)	$\mathbf{a}^*$ (averaged over α_i, β_{ji})							
$a_0 = 5$	5.35	5.	5.	5.	4.98	5.	5.	5.
$a_1 = 6$	5.85	5.93	5.95	5.90	5.57	5.83	5.94	5.81
$a_2 = -3$	-2.61	-2.86	-2.97	-2.91	-2.58	-2.84	-2.96	-2.83
$a_3 = -1$	-1.13	-1.08	-1.06	-1.05	-1.07	-1.05	-1.04	-1.04
Variance $h(\mathbf{a}) = \max \mathbf{a} - \min \mathbf{a}$								
$h(a_0)$	0.88	0.01	0.04	0.02	0.05	0.00	0.00	0.00
$h(a_1)$	1.43	0.18	0.13	0.18	0.02	0.09	0.06	0.05
$h(a_2)$	0.46	0.11	0.12	0.11	0.12	0.16	0.11	0.16
$h(a_3)$	1.3	0.05	0.04	0.05	0.03	0.05	0.03	0.02

($\mathbf{a}^{LS}$ evaluations: $a_0 = 5.$, $a_1 = 5.81$, $a_2 = -2.83$, $a_3 = -1.05$).

Table 2 demonstrates the desired stability of the optimal $\mathbf{a}^*$ with respect to outliers (points no. 4,6,8-10) when using operator MA ($\mathbf{a}^{LS}$ turned out to be heavily biased).

Table 2. Detection of outliers

	a_0	a_1	a_2	a_3	$MA(i) = MA_i(\mathbf{a}^*)$
$\mathbf{a}^*$	5	6	-3	-1	
$\mathbf{a}^{LS}$	7.55	3.50	-2.84	-1.03	
$\mathbf{a}^*$					
$m = 1.2$	5.79	7.44	-2.95	0.35	$MA(4,10)<0.01$, $MA(6,8)<0.001$, $MA(9)<0.1$
$m = 2$	5.14	5.92	-2.71	-1.12	$MA(4,6,8,9,10)<10^{-6}$
$m = 4$	5.02	5.96	-2.97	-1.07	$MA(4,6,8,9,10)<10^{-36}$
$m = 10$	4.99	5.99	-3.00	-1.04	$MA(4,6,8,9,10)=0$

Summarizing we obtained the following results.

1. $\mathbf{a}_A^*$ and $\mathbf{a}_G^*$ differ only little if disturbance of data is small.
2. They are stable with respect to α and β .
3. Same for m . Preferably $m = 4, 10$.
4. Estimations for $\mathbf{a}_A^*$ were stable w.r.t. outliers if their number was less than 30%.
5. Stability for various disturbances of data.
6. For weakly correlated inputs ($r < 0.5$) estimations are nearly unbiased. For $r > 0.9$ the results became useless.

6. Technical Application

Now let us return to the problem formulated in Section 2. In Table 3 we give the technical characteristics for *CL*. Analysis shows that the fuzziness of data can be modelled by (10) with $m = 10, s = 1$.

Table 3. Technical data

N	concentration of emulsion [%] x_1	biological impairment x_2	corrosivity y
1	5.2	0	0
2	3.0	0	1
3	1.5	0	1
4	1.9	0	1
5	3.4	1	1
6	3.9	2	2
7	4.1	2	3
8	3.3	3	3
9	3.2	4	4
10	8.0	3	0
11	6.9	3	0
12	7.8	4	1
13	2.1	1	4

We took $\alpha_i = \alpha = 1$; $\beta_{1i} = \beta_1 = 1\%$; $\beta_{2i} = \beta_2 = 1$. As the result of computation using $MA(\mathbf{a})$ we got

$$y = 3.53 - 0.89 x_1 + 1.11 x_2$$

with $MA(\mathbf{a}^*) = 0.999$. Now for the task

$$x_1 \rightarrow \min$$

$$0 \leq x_2 \leq 4$$

$$y \leq 1$$

we obtained $x_1 \geq 7.83 \%$.

References

1. D. Dubois, H. Prade, Fuzzy Sets and Systems - Theory and Applications. Academic Press, N.Y., 1980.
2. H. Bandemer, S. Gottwald, Einführung in FUZZY-Methoden (in German). Akademie-Verlag, Berlin, 1989.
3. D. A. Pospelov, Fuzzy Sets in Models of Control and Artificial Intelligence (in Russian). Nauka, Moscow, 1986.
4. D. Himmelblau, Applied Nonlinear Programming. McGraw-Hill, N.Y., 1972.
5. K. Cicinadze, Solution of Nonconvex and Nonlinear Optimization Problems (in Russian). Nauka, Moscow, 1983.